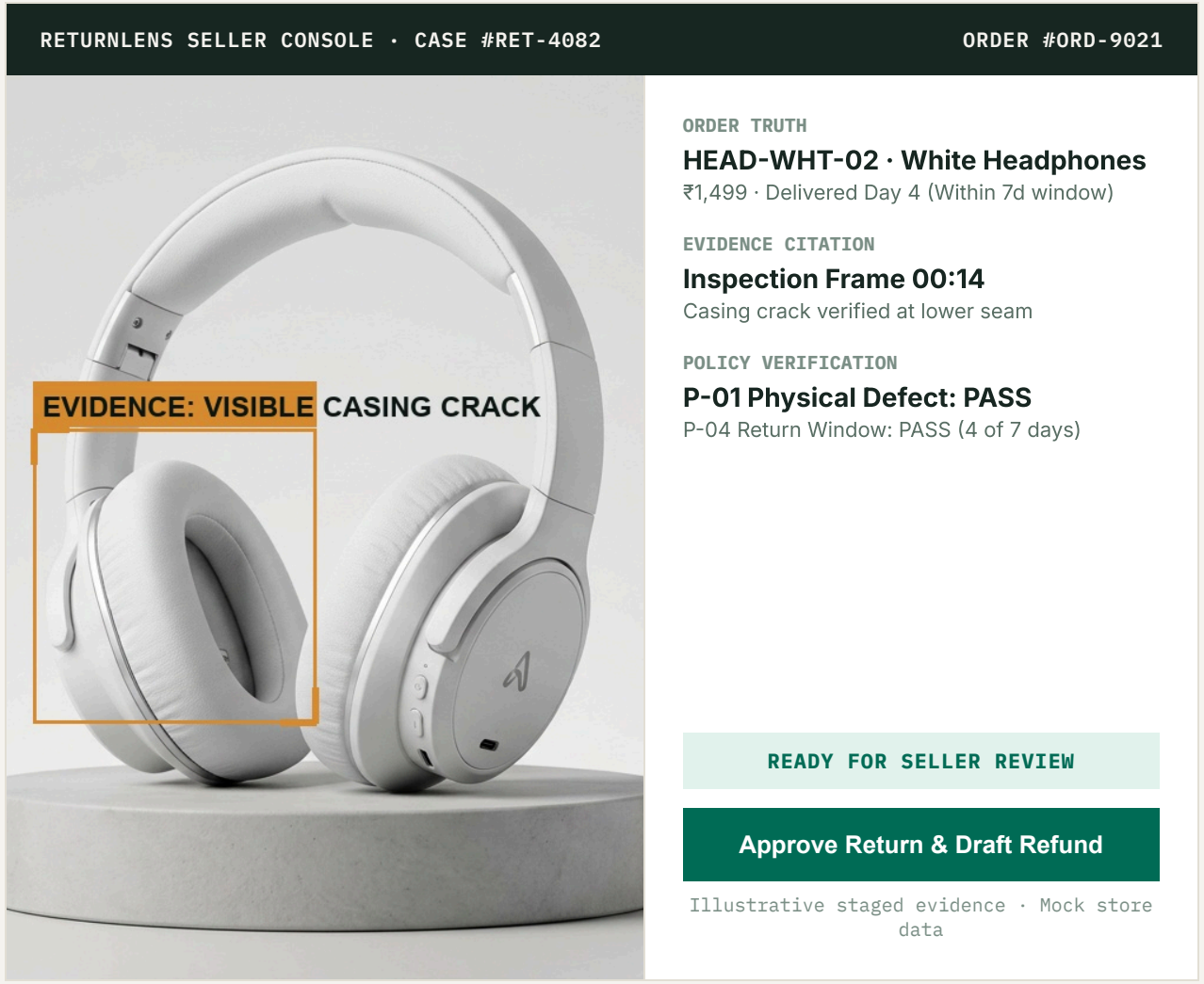
AGENTX HACKATHON 2026 · TRACK PS-04

RETURNLENS

“Review returns with cited video evidence.”

An evidence-backed return triage agent for small direct-to-consumer sellers.



Team Rookie · 3 members · PS-04 submission

User: Solo D2C merchant managing orders and returns alone.
Scope in 24h: Return triage only, mock store data + 1 order-status route.
Not another chatbot; not an autonomous refund gate.

THE PROBLEM · SOLO SELLER BOTTLENECK

One return decision. Four disconnected sources.

The return decision is made from memory, before the evidence is checked.

01 BUYER CHAT

"Received damaged item"

Buyer upload: 30–60s unboxing video + reason

02 ORDER RECORD

White Headphones · ₹1,499

tabase

CRITICAL INFORMATION GAP

NO SINGLE RECORD TIES ORDER, VIDEO & POLICY TOGETHER

Today the seller opens many tabs, then decides alone. A wrong refund loses money and stock; an unfair denial loses a repeat customer. The gap is not a missing returns portal, it is that no single record ties order, video frame, policy clause, and decision together.

03 COURIER TRACKING

Delivered 4 days ago

Shippo mock tracking API timestamp verification

7-Day Window · Sealed Tag

Deterministic code rules P-01 to P-08, never prompts

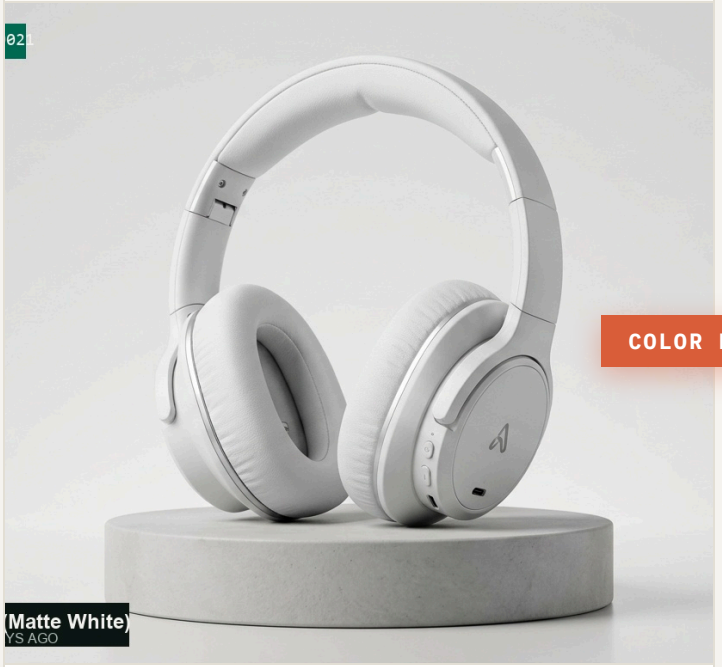
Fraud estimates differ by definition: 9% (NRF 2025, fraud only) vs 15.14% (Appriss/Deloitte 2024, fraud + abuse).

VISUAL EVIDENCE BENCHMARK · CORE DEMONSTRATION

The order says white. The video shows black.

ReturnLens surfaces the contradiction instead of auto-denying.

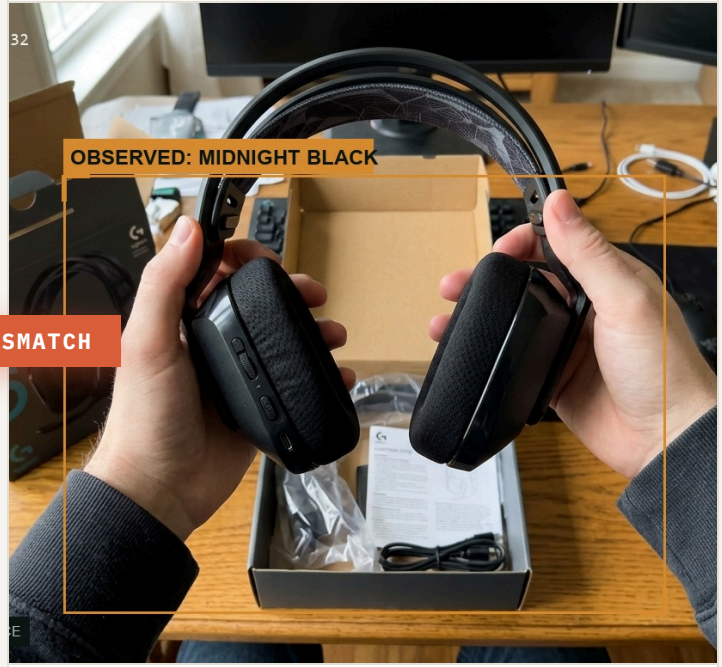
ORDER RECORD · STORE GROUND TRUTH



HEAD-WHT-02 · Matte White · ₹1,499
Delivered 4 days ago · Original tag present

COLOR MISMATCH

VIDEO FRAME · INTAKE TIMESTAMP 00:07



Observed: HEAD-BLK-09 · Midnight Black
Mock evidence pack · Frame 00:07

DETERMINISTIC POLICY GATE

Policy P-03 Check: FAIL

Specification contradiction flagged against Shopify order manifest.

```
typed JSON, requires_seller_approval: true
```

STATUS: ESCALATE (NEEDS SELLER REVIEW)
Model proposes; human authorizes. No label or refund auto-issued.

SELLER ACTION OPTIONS

1. Accept as merchant mis-shipment (seller mistake)
2. Ask customer for outside-box shipping label (buyer mistake)
3. Hold return for physical inspection at warehouse (potential swap)

Deterministic Guardrails:
- Cap: 0 < amount ≤ min(₹4,999, remaining refundable amount, order paid amount)
- No automatic customer denial · No refund released without seller click

SYSTEM ARCHITECTURE · DETERMINISTIC PIPELINE

Models observe. Code verifies. Seller authorizes.

Policy checks and seller approval control actions.

01 PLAN	02 RETRIEVE	03 EVALUATE	04 AUDIT	05 AUTHORIZE	06 RECORD
Planner	**Tools & Video**	**Policy Engine**	**Critic**	**Seller Gate**	**Audit Log**
MODEL	CODE	CODE	MODEL	HUMAN	CODE

1. CONTRADICTION → ESCALATE

Discrepancy flagged (e.g. Case B). Inhibits automated label and refund draft. Seller reviews 3 options.

2. HARD FAIL → ESCALATE

Mandatory rule violation (e.g. expired return window). Seller makes final decision; never auto-deny.

3. UNCLEAR ONLY → ASK MORE

Low brightness or blurry barcode (e.g. Case C). Prompts exact re-film checklist (1 retry limit).

4. ALL PASS → APPROVE REVIEW

All clauses pass (e.g. Case A). Prepares mock label and drafts refund in pending_return_receipt.

- **Model Calls:** Exactly 3 base calls per case: planner, vision, critic, plus retries
- **Policy Clauses:** Deterministic code rules P-01 to P-08, never prompts
- **Critic:** The critic reviews cited frames and flags unsupported observations
- **Safety Limit:** Injection cannot alter policy code or bypass human gate

DEMO SCENARIOS · GROUNDED VERIFICATION

Four scenarios. Four predictable behaviors.

Every decision can be clicked back to a frame, a clause, and a log entry.

CASE A · READY	**ORD-1042** (₹1,499) · SKU matches manifest, tag present, visible defect within 7-day window.	→	Seller approval releases mock return label and refund draft in **pending_return_receipt**, not executed refund.
CASE B · ESCALATE	**HEAD-WHT-02** vs **HEAD-BLK-09** at Frame 00:07 · Policy P-03 SKU mismatch **FAIL**.	→	**ESCALATE** (no label, no refund, no denial). Seller options: mis-shipment / box label / physical inspection.
CASE C · ASK MORE	**4-second dark video**, brightness 28, only 2 usable frames · Unclear evidence check.	→	Case stays open with exact re-film checklist. One retry limit prevents customer lock-out.
CASE D · CONTINUITY	**Unbroken take**: opening sealed parcel to showing the label in one continuous sequence.	→	Continuity verified: prevents photo-swap fraud before mock return label generation. Proves why video beats photos.

EXECUTION PLAN · TEAM ROOKIE (3 MEMBERS)

Built for a 24-hour sprint.

Buildable in 24 hours by 3 people, if the scope stays narrow.

PERSON A

Backend & Agent Core

Backend, agent graph, policy, critic, audit log.

PERSON B

Video Pipeline & UI

Video pipeline, video indexer, seller console, evidence pack view.

PERSON C

Integration & Verification

Integration, mock data, 4 demo cases, fallback testing, demo video, README.

NEVER CUT (CORE ENGINE)

Deterministic policy engine (P-01 to P-08) · Evidence-pack frame citation · Fail-closed seller approval gate · Structured JSON audit log · 4 reproducible demo cases.

CUT FIRST (DEFERRED EXPANSION)

Video OCR barcode scanner · Auto-drafted customer email reply · Seller analytics dashboard · Production payment gateways / live WhatsApp webhooks.

"The winning hackathon agent is not the one with the most tool definitions. It is the one where every action is held to evidence, and the seller stays in control."